

Project Report: PantryChef

AI-Powered Recipe Generation System

Course: MLOps

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Abstract

PantryChef is an intelligent web application designed to reduce food waste and decision fatigue by generating personalized recipes based on available ingredients. The system leverages Large Language Models (LLMs) via Hugging Face Space, specifically utilizing the **Qwen2.5-7B-Instruct** model to convert unstructured ingredient lists into structured, actionable recipes.

Built using **Next.js 16** and **React 19**, the application ensures a seamless user experience with features such as ingredient autocomplete, dietary profiling, and recipe history. The backend architecture integrates **Supabase** for robust authentication and data persistence, while the AI inference is decoupled via a **Gradio API** deployed on **Hugging Face Spaces**. This report details the system architecture, implementation strategy, and initial results of the project, demonstrating a practical application of MLOps principles in a consumer-facing product.

1. Introduction

1.1 Background and Motivation

Cooking at home offers nutritional and financial benefits, yet many individuals struggle with "pantry paralysis", the inability to decide what to cook with the ingredients on hand. Traditional recipe platforms require users to search for specific dish names, often leading to unnecessary grocery purchases and increased food waste.

The emergence of Large Language Models (LLMs) allows for a semantic understanding of ingredients. By treating recipe creation as a constrained generation problem, we can use AI to bridge the gap between a user's current inventory and a delicious meal.

1.2 Problem Statement

The project addresses four specific challenges:

- **Ingredient Constraints:** Users need recipes based strictly on what they own.
- **Dietary Complexity:** Recipes must dynamically adapt to allergies (e.g., Nut-free) and preferences (e.g., Keto, Vegan).
- **Structured Data:** LLMs naturally output text; the challenge is forcing structured JSON output for a functional UI.
- **Latency & Scale:** The system requires an architecture that separates the heavy AI inference from the user interface.

1.3 Objectives

1. **AI Integration:** Deploy an LLM (Qwen2.5-7B) capable of following complex instruction prompts to generate JSON recipes.
2. **User Experience:** Develop a user-friendly interface with "Smart Pantry" autocomplete functionality.
3. **Persistence:** Implement a database schema to store user "Food DNA" (profiles) and recipe history.
4. **MLOps Loop:** Establish a feedback mechanism (star ratings) to collect data for future model fine-tuning.

2. Proposed Methodology

2.1 System Architecture

PantryChef utilizes a decoupled Client-Server architecture. The Next.js frontend handles user interaction, while the heavy lifting of AI inference is offloaded to a dedicated Python environment on Hugging Face Spaces.

Data Flow:

1. **Input:** User inputs ingredients and selects preferences on the Frontend.

- 2. **Orchestration:** Next.js API Route authenticates the request via Supabase.
- 3. **Inference:** The request is forwarded to the Gradio Client.
- 4. **Generation:** Qwen2.5-7B processes the prompt on Hugging Face Spaces.
- 5. **Storage:** The resulting recipe is parsed and stored in Supabase.

2.2 Technology Stack

Component	Technology Used	Purpose
Frontend	Next.js 16 (App Router)	Main application framework and API routing.
UI Library	React 19, Tailwind CSS 4	Component architecture and styling.
Database	Supabase (PostgreSQL)	User profiles, recipe storage, and Auth.
AI Model	Qwen2.5-7B-Instruct	Instruction-tuned LLM for recipe generation.
Model Hosting	Hugging Face Spaces	Containerized environment for running the model.
Interface	Gradio SDK	API bridge between Next.js and the Python model.

2.3 Prompt Engineering Strategy

To ensure the LLM outputs data usable by our application, we implemented a strict **System Prompt**. The prompt enforces:

1. **Role:** "You are a professional chef."
2. **Constraint:** "Use ONLY provided ingredients (plus basic staples), and follow user preferences for type of cuisine, allergies."
3. **Format:** "Output valid JSON only. No markdown, no conversational text."

This strategy transforms the probabilistic nature of the LLM into a deterministic data source for our frontend.

3. Implementation & Results

3.1 Implementation Status

The core functionalities of the system have been fully implemented and deployed.

- **Authentication:** Users can sign up and log in via Supabase.
- **AI Generation:** The Qwen2.5 model successfully generates recipes in <10 seconds.
- **JSON Parsing:** A robust regex parser cleans the LLM output to prevent UI crashes.
- **Mobile Responsiveness:** The interface includes touch-optimized inputs and sticky navigation.

3.2 Sample Output

The following is a raw JSON response generated by the system given the inputs *["Chicken", "Rice", "Tomatoes"]*:

JSON:

```
{
  "title": "Spicy Tomato Chicken & Rice",
  "description": "A hearty one-pot meal perfect for weeknight dinners.",
  "ingredients": [
    "500g Chicken breast, cubed",
    "2 cups Rice, rinsed",
    "3 Tomatoes, diced",
    "1 tbsp Oil (Staple)",
    "1 tsp Salt (Staple)"
  ],
  "instructions": [
```

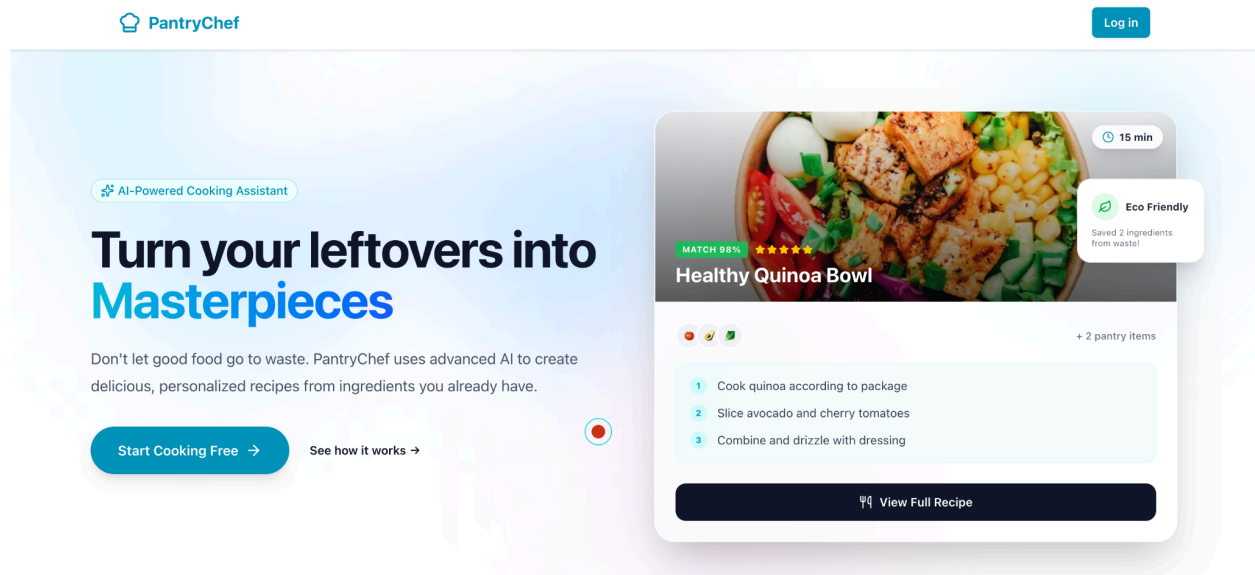
```

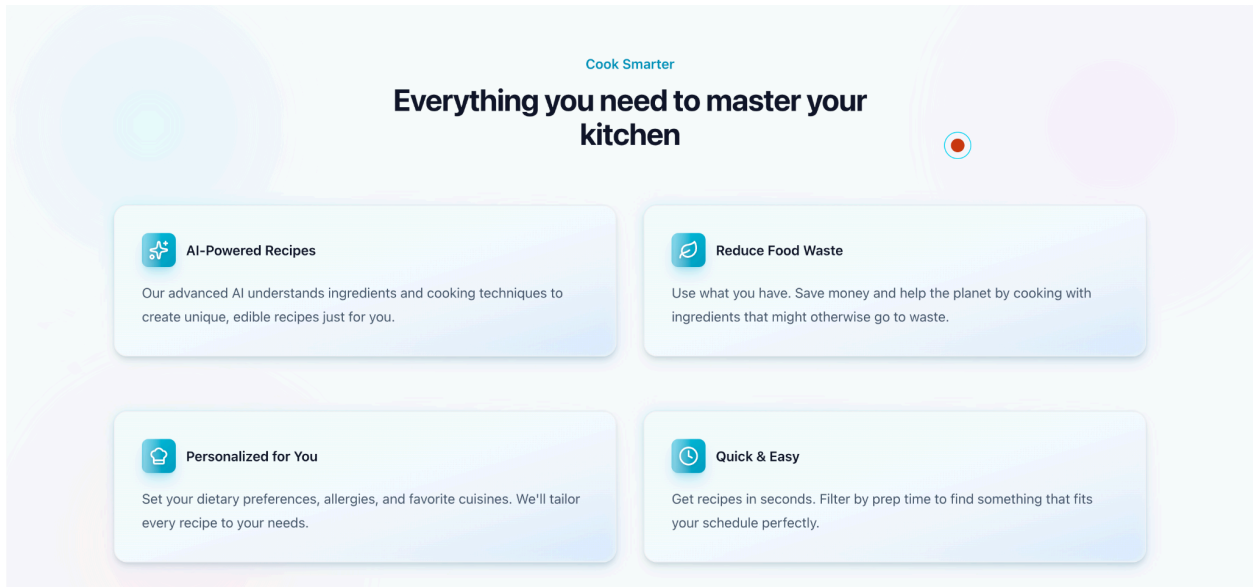
    "Heat oil in a large pan.",
    "Sauté chicken until golden.",
    "Add tomatoes and cook until soft.",
    "Stir in rice and add water.",
    "Cover and simmer for 20 minutes."
  ],
  "prep_time": "35 mins"
}

```

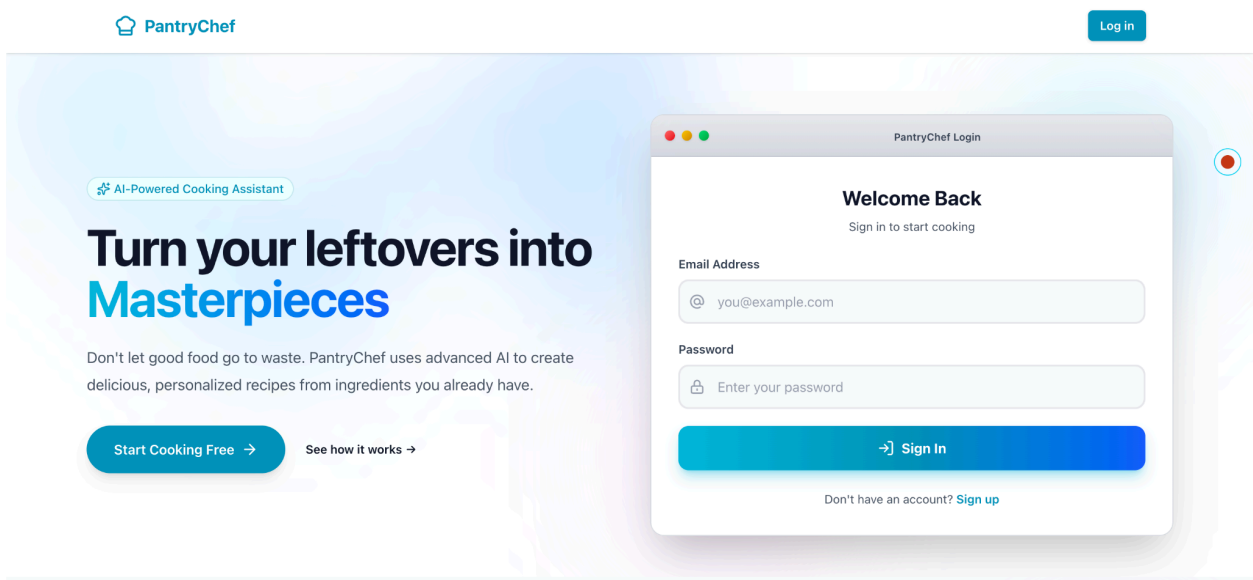
3.3 User Interface

Landing Page:

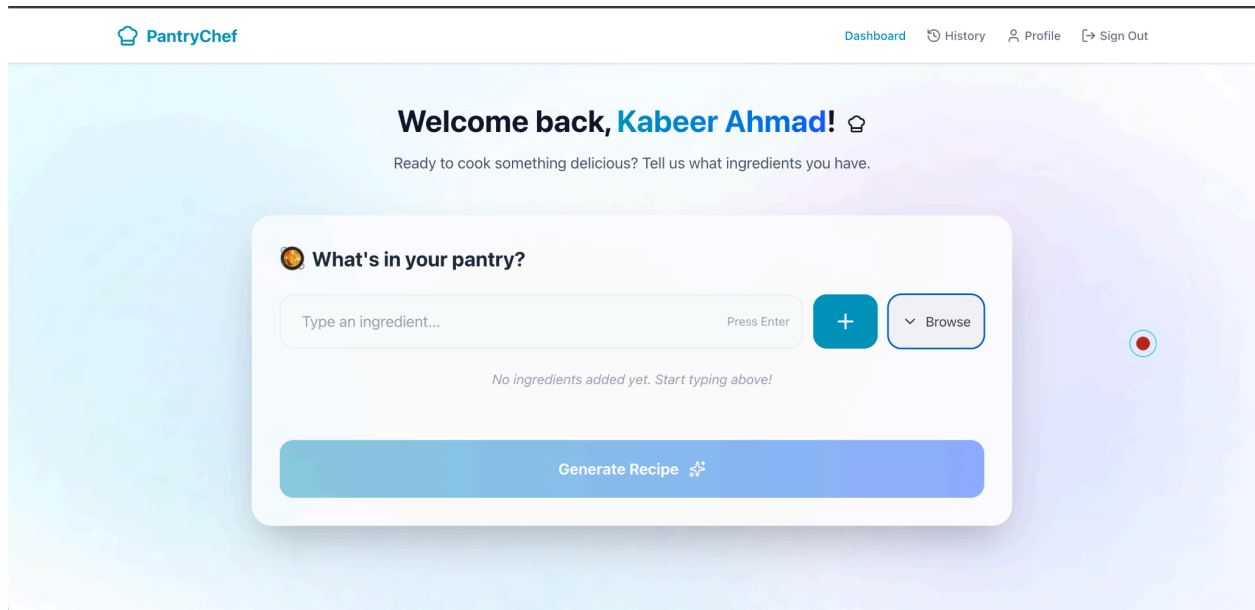




Login:

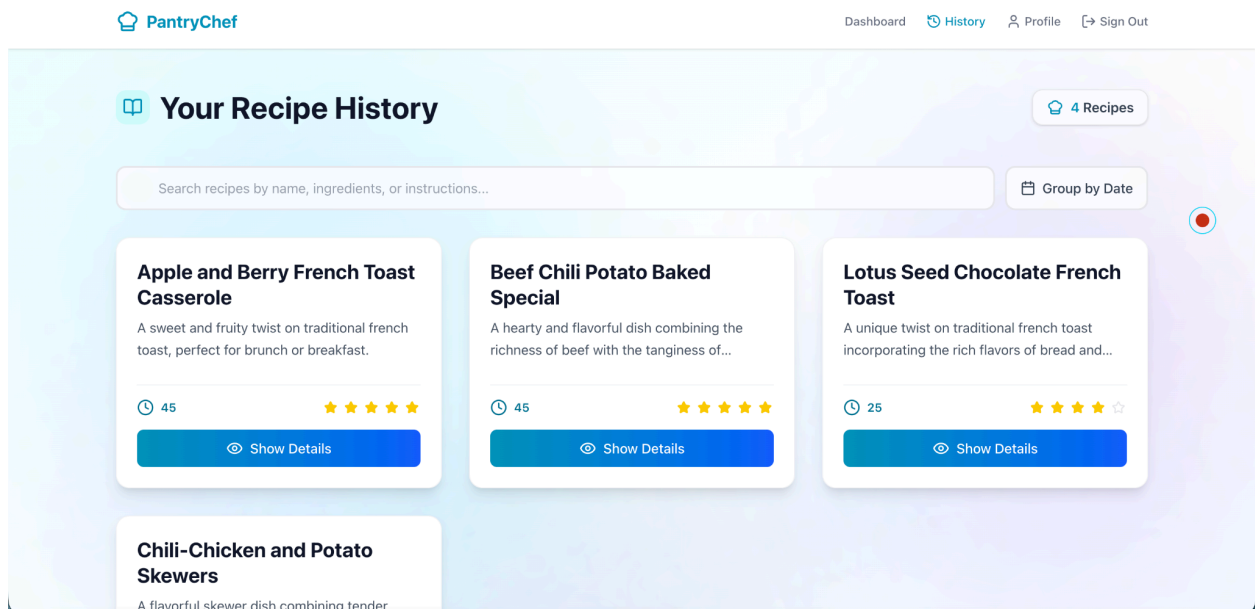


Dashboard:



The main interface allows users to select ingredients via an autocomplete search bar or category browser. The design features a "Glassmorphism" aesthetic with animated gradients.

History:



Profile Page:

PantryChef

DashboardHistoryProfileSign Out



Your Profile
Customize your cooking preferences

Display Name
Kabeer Ahmad

**Quick Tip**
Leave these blank if you have no specific restrictions. We'll assume you enjoy everything!

**Dietary Preferences**

Vegan

Vegetarian

Pescatarian

Gluten-Free

Keto

Paleo

Dairy-Free

Halal

Kosher

Low Carb

Whole30

Low FODMAP

**Allergies**

Peanuts

Tree Nuts

Milk

Eggs

Wheat

Soy

Fish

Shellfish

Sesame

Mustard

Sulfites

Corn

Other Allergies

Other Allergies

Strawberries, Latex...

**Favorite Cuisines**

Pakistani

Italian

Mexican

Chinese

Japanese

Indian

Thai

French

Mediterranean

American

Korean

Vietnamese

Greek

Turkish

Lebanese

Spanish

Save Changes

Demo Link:

<https://drive.google.com/file/d/1yzIt5-rHGi-QpC9y9U0ENYB78bOq5K3C/view?usp=sharing>

4. Future Improvements & Limitations

4.1 Current Limitations

- **Hallucinations:** Very rarely, the model may suggest an ingredient not in the user's list (hallucination).
- **Latency:** Cold starts on the Hugging Face Space can lead to a 10-15 second delay for the first request.

4.2 Future Roadmap

1. **Image Generation:** Integrate Stable Diffusion to generate realistic images of the final dish based on the recipe title.
2. **Shopping List:** Allow users to "fill the gaps" by generating a shopping list for recipes they *almost* have the ingredients for.
3. **Fine-Tuning:** Use the collected 1-5 star rating data to fine-tune a smaller model (e.g., Llama-3-8B) specifically for this application to improve consistency.

5. Conclusion

PantryChef successfully demonstrates the capabilities of modern AI in solving daily problems. By combining a robust full-stack framework (Next.js/Supabase) with powerful open-source AI (Qwen/Hugging Face), we have created a tool that is not only technically sound but also practically useful. The project highlights the importance of **Prompt Engineering** and **System Architecture** in building reliable AI-powered applications.